

Joel Teply

Principal / Staff Engineer — Computer Vision, ML Systems, High-Performance Computing

Kansas City Metro · joel@cambriantech.com · github.com/CambrianTech

PROFILE

Twenty years building computer vision, machine learning, and high-performance systems — from pre-ARKit mobile scene understanding and biometric perception to LLM architecture and inference runtimes. I make research-grade computation practical on constrained hardware: models, runtimes, GPU kernels, memory systems, schedulers, and the product above them.

The work includes patented computer vision and liveness detection, real-time 3D scene understanding on iPhone-class hardware deployed to millions of users, and current research in attention-head plasticity, structured pruning, and continual learning.

The through-line is the same problem in different decades: take something that assumes a datacenter, a research rig, or specialized silicon, and make it work on the machine someone already owns. Mobile was usually the deployment target, not the discipline — phones were the constrained computers available at scale.

EXPERIENCE

Cambrian Tech — Co-founder, CTO, Principal Engineer · 2009–present

Co-founded a computer vision and augmented reality technology company building proprietary perception technology and delivering custom engineering programs for enterprise clients across automotive, retail, and home improvement. Owned technical architecture and implementation end to end — ML, computer vision, real-time 3D perception, GPU optimization, native and web platforms — setting technical direction and working with engineering teams and customers to turn hard perception problems into deployable systems. Technology licensed to Fortune 500 customers including Home Depot and Lowe's, deployed to millions of users.

Automotive computer vision — Garmin, 2020–2022. Led the perception and spatial-computing engineering for an automotive program delivered under strict accuracy and CPU/GPU constraints, from research through optimization and delivery; successful delivery expanded into a multi-year engagement.

Perception and spatial computing (2009–2022)

- **Real-time asynchronous perception.** Maintained a continuously evolving scene representation at frame rate while expensive CNN inference ran asynchronously on-device with 1–3 second latency. Cellular-automaton region evolution over learned patch affinities, watershed segmentation across detected structural lines, tracking and temporal interpolation continuously added, removed, extended and joined regions; delayed semantic-segmentation and surface-normal results revised the live state as they arrived rather than gating the frame rate. Segmentation and surface normals both ran on the device — no cloud.
- **Two kinds of consistency, held at once.** Geometric accumulation and tracking gave spatial consistency — optical flow, camera pose, RANSAC plane fitting, multi-frame fusion, SLAM-like structure from an ordinary camera. Asynchronous state revision gave perceptual consistency despite slow inference. Camera parameter estimation (field of view, rotation), panorama-to-pinhole reprojection, and online calibration in C++.
- **The pipeline that made it possible.** Priority-scheduled and backpressured C++ perception pipeline over lazily evaluated shared frame representations, so estimators with radically different cost ran at their own cadence without stalling the world model. Optimized with SIMD, threading, and OpenGL ES compute/GPGPU shaders. One core architecture targeting iOS/Objective-C++, Android/JNI, Unity, browser (TensorFlow.js, WebGL), and headless servers.

- Walkable AR flooring **before ARKit existed**: a rendered floor that stays locked to the room as you walk on it, on large low-texture foreshortened surfaces where plane fitting fails and only real-time semantics hold the boundary.
- **Photographic compositing rather than overlay**. Scene lighting and shadow estimation, texture and material decomposition, and re-rendering surfaces under the room's real illumination. The realism bar came from a visual-arts background — knowing what makes a composite read as a photograph rather than a sticker drove which physical quantities the perception stack had to recover in the first place. Shipped in Wall Painter and Home Harmony (stills) and **Video Painter (App Store, 2011)** in live video, when mobile AR still meant sprite overlays.
- First-named inventor, **US10964097B2** — *Pattern Recognition Systems and Methods for Performing Segmentation on Surfaces and Objects*.

Applied ML research (2009–2022)

- Learned-affinity segmentation **before AlexNet**: a modified Siamese network learned the patch-similarity metric that drove the region evolution above. On our task it outperformed ImageNet-pretrained semantic segmentation and ran far faster, being adaptive to the image and evaluating a growing frontier rather than dense inference. Migrated to CNNs once they generalized without a seed and had hardware acceleration behind them.
- Weight quantization and model compression tooling for deployed CNNs (2017) — the same problem now being solved at LLM scale.
- ICNet retargeted for both semantic segmentation and surface-normal prediction, trained and exported for mobile inference; soft segmentation and matting for boundary quality.
- Cross-framework model engineering: Caffe to TensorFlow and Caffe to CoreML/MLKit conversion so one trained model shipped to every target.
- Synthetic training data: image synthesis pipelines, including Unreal Engine scenes, built to produce labeled ground truth where real data was unobtainable.
- Generative work: inpainting and seamless texture synthesis for surface replacement; photo-driven video generation.

LLM systems and continual learning (2023–present) — see Selected Projects.

TripleBlind → Ideem — Software Engineer, Cybersecurity · 2023–2026

Cryptographic authentication technology: device-bound key handling without dedicated secure hardware, secure client integration, and the SDK surface enterprise customers build against.

EyeVerify / ZOLOZ — Software Engineer, Computer Vision & Biometrics · 2012–2016

One of the first two engineers on eye-vein biometric authentication; acquired by Ant Financial (Alibaba) for over \$100M in 2016.

- Identified scleral vasculature from commodity front-facing cameras with verification in under a second; eye and iris tracking, segmentation, multi-frame fusion, and image enhancement at camera frame rate via OpenGL/GLSL and threaded pipelines.
- Presentation-attack and liveness detection: distinguishing a live person from a photo, replay, or mask. Co-inventor, **US9665784B2** — spoof detection using only a phone's own earpiece and microphone (sonic 3D face sensing, photometric analysis, multi-source pulse detection). Also named on **US8437513B1** and **US8675925B2**.
- Turned research algorithms into a commercial SDK; stayed through acquisition and integration.

Earlier

Propaganda3 — Software Engineer · 2010–2012. Mobile and web systems including a multi-state tax application with photo-based W-2 ingestion. **VML** — Software Engineer · 2006–2010. Backend, web, device, and early mobile systems. **AgileWise** — Hardware & Software Engineer · 2005–2006. PCB design, a touchscreen embedded-computer prototype, RFID integration, Windows CE platform image and display drivers. **NovaStar · Federal Home Loan Bank**

of Topeka · AllofE Solutions · 1999–2006. Backend and B2B systems; earlier, PCB assembly for aerospace and government customers.

SELECTED PROJECTS

Continuum — open-source Rust platform for persistent AI agents running across commodity hardware. Sole architect, 550,000 lines. Multi-agent orchestration with governed tool execution; typed KV-slot leases and priced eviction so a dozen agents share a few warm slots; evaluation-gated continual learning where agent experience generates LoRA curricula, reaching 43 of 76 resolved SWE-bench Verified attempts (57%) with reproducible verdict artifacts; peer-to-peer distribution of inference, training and adapters with signed provenance. 121 commits on a maintained llama.cpp fork covering backend lifetime, KV-cache behaviour, and Metal/CUDA placement. Sustains ~50 tok/s solo and ~128 batched on Metal, 80–237 on consumer CUDA; renders fourteen concurrent embodied agents at 30 fps on the same machine serving their models. github.com/CambrianTech/continuum

Sentinel-AI — transformer architecture research with a published paper, *Experiential Plasticity*: attention heads that contribute to a domain specialize while the rest are removed, so architecture co-evolves with training. Qwen3.5-4B forged on code improved perplexity 24% over baseline from a smaller model; the 27B improved 3.5% and runs in 17 GB at 4-bit against 28 GB at fp16. Models published on HuggingFace with their recipes.

github.com/CambrianTech/sentinel-ai

Model compaction — head pruning by gate-gradient utilization took a 14B coder model from 27 GB to 8.9 GB; runtime activation profiling pruned a 35B mixture-of-experts from 67 GB to 47 GB. Published with reproducible recipes.

TECHNICAL

ML architecture and research — metric learning and Siamese networks; CNN architectures for segmentation, depth, surface normals, classification and reconstruction; GANs, diffusion and generative pipelines; synthetic data; transformers and LLMs; attention-head plasticity, LoRA, structured pruning, quantization, and inference-runtime engineering.

Perception and spatial computing — semantic segmentation, monocular depth, optical flow, camera pose and calibration, plane fitting, multi-frame fusion, scene reconstruction, SLAM-like systems, tracking, biometrics, real-time AR compositing.

Systems and performance — Rust, C++, Python, Go, TypeScript, C#; SIMD, concurrency, GPU/GPGPU (Metal, CUDA, OpenGL/GLSL, WebGL), memory-constrained execution, scheduling and backpressure, distributed systems, embedded and edge platforms, structured telemetry.

Platforms — iOS/Objective-C++, Android/JNI, Linux, Unity, Unreal Engine, browser ML, native camera pipelines, embedded and PCB-level hardware.

EDUCATION & PATENTS

B.S. Computer Engineering — University of Kansas

- **US10964097B2** — *Pattern Recognition Systems and Methods for Performing Segmentation on Surfaces and Objects* (2021, Cambrian Tech LLC). First-named inventor.
- **US9665784B2** — *Systems and Methods for Spoof Detection and Liveness Analysis* (2017).
- **US8437513B1**, **US8675925B2** — *Spoof Detection for Biometric Authentication* (2013, 2014).
- \$100,000 LaunchKC grand prize (2017); top-five finalist, Atlanta Startup Battle 6.0 (2019, TechSquare Labs).